Unidentified Bright Objects (UBOs)

UBOs may be found in the internal capsule, basal ganglia, cortex, cerebellar hemispheres, optic tract, or brainstem. They do not enhance and are not associated with compression of surrounding tissue, features that distinguish them from neoplasms. UBOs are present in approximately 60 percent of children with NF-1 but disappear with age and are uncommon in adults. They are not associated with focal neurologic signs and are of uncertain significance. Histopathologic studies have shown that the UBOs correspond to areas of myelin vacuolization with increased water content. Some studies have found that children with NF-1 who have UBOs have significantly lower IQ and language scores, show significantly impaired visual-motor integration and motor coordination, and are at much greater risk for impaired academic achievement compared with NF-1 patients without UBOs.

Intelligence and Learning Disabilities

Contrary to gross misstatements in the older literature, the mean IQ of patients with NF-1 is only 5 to 10 points lower than that of the general population and unaffected siblings. However, learning disabilities, defined as discrepancies between ability (intellect) and performance, are quite common, with an estimated prevalence of 30 percent to 60 percent. Although earlier reports suggested there might be a specific "cognitive phenotype" in NF-1 characterized by an excess of visual-perceptual disabilities, more recent studies have demonstrated that verbal deficits (e.g., reading) are at least as common as nonverbal disabilities in individuals with NF-1. These learning disabilities are lifelong and can affect adult functioning. Many children with NF-1 also have poor attention and impulse control and may be diagnosed with attention-deficit/hyperactivity disorder, which significantly interferes with school performance and learning. The use of stimulant medications may have a profound beneficial effect on the ability of these individuals to succeed academically and professionally.

Treatment of Neurofibromatosis Type 1

Individuals with NF-1 are best cared for within a multidisciplinary clinic that has access to a wide range of subspecialists. The exact physician composition of the clinic is less important than the ability to obtain expeditious subspecialty consultation. All first-degree relatives should be examined for the cutaneous manifestations of NF-1 and should undergo slit-lamp examination at the first visit to ascertain the presence of Lisch nodules. Yearly visits allow the physician to identify NF-1 complications early while providing counseling and dissemination of information regarding NF-1. Individuals and families can obtain further information from the websites of the two national support groups, the Children's Tumor Foundation (http://www.ctf.org) and Neurofibromatosis, Inc. (http://www.nfinc.org).

All children with NF-1 who are 10 years of age or younger should have yearly complete ophthalmologic examinations to look for signs of an optic pathway tumor (OPT). These examinations should include assessment of visual acuity, color vision, and visual fields, as well as funduscopy and slit-lamp examination. Because almost all OPTs arise in children in this age group, the frequency of ophthalmologic examinations can be reduced in children older than 10 years of age. Yearly measurements of weight and height should be plotted on standardized growth charts, because the earliest indication of precocious puberty may be accelerated linear growth. Blood pressure measurements should be obtained at each visit to look for signs of renovascular hypertension. In addition, the spine should be examined each year for early signs of scoliosis.

Café-au-Lait Spots

Café-au-lait spots on the face are unusual in individuals with NF-1. On occasion, they do occur, and affected individuals may seek to improve cosmesis. Attempts to remove café-au-lait spots with laser therapy have provided very mixed results. Although total disappearance of the lesion may be achieved, some studies cite a recurrence rate as high as 67 percent. Although results with use of the Q-switched ruby laser (694 nm) have been mixed, favorable reports in a limited number of cases using either the continuous-mode copper vapor laser (511 nm) or the erbium:yttrium-aluminum-garnet laser (2940 nm) have been published. Multiple treatments of a single lesion may be more effective.

Cutaneous Neurofibromas

Discrete cutaneous neurofibromas may be removed surgically to improve cosmesis or to prevent local irritation (e.g., from brushing for lesions in the hairline or from rubbing against the shoe for those on the foot).